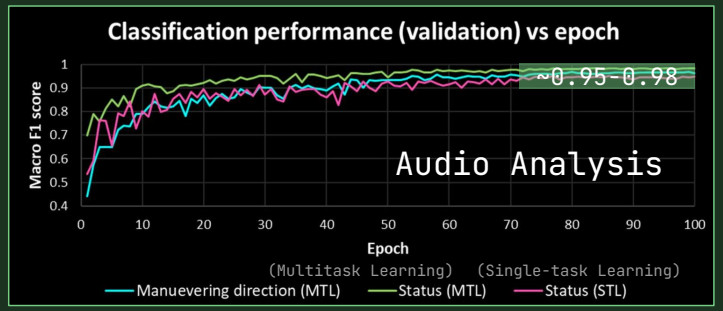
Verification & Validation

1 Fault Diagnosis

$$\text{Precision (P)} = \frac{\text{True Pred +ve}}{\text{All Pred +ve}}$$

$$\text{Recall (R)} = \frac{\text{True Pred +ve}}{\text{All Real +ve}}$$

F1 Score: Harmonic Mean of Precision and Recall



Audio analysis deep learning model performance on the research paper.
We achieved 0.974 F1 score on the dataset audio.

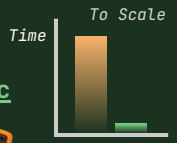
Audio Analysis Our Drone Audio	Blackbox Analysis Our Drone Flight Data
Accuracy: 0.815/1 F1 Score: 0.715/1	Accuracy: 0.868/1 F1 Score: 0.929/1

2 Diagnostic Reliability

Tested under noisy and quiet environments to simulate varying conditions, and maintained consistent results.

3 Operational Integration

Betaflight website serial log extraction: ~5 min
Mass storage mode extraction from device: <30 sec



4 Field Portability

Size of handheld multimeter, runs off power bank, stores data locally

5 Diagnostic Traceability

Every test was checked for logging completeness; MicroSD with ~100GB storage; Data synced to cloud database

Validated workflow through real-world testing by non-technical users with no intervention required.

Sustainability: Individual components are easily replaceable, code can be open-sourced for customization

Future Work

- **More Testing:** Test across D. Vision Aerials' full drone fleet, including windy and cold conditions.
- **Deeper Analysis:** Expand black-box analytics (battery monitoring, PID tuning) and improve algorithm speed.
- **Enterprise System:** Add user accounts for fleet-wide diagnostic tracking.
- **Additional Sensors:** Extend diagnostics with more sensors (e.g., thermal monitoring).

Team 15

Javier Huang
Shupeng Liu
Yangxin Zhou
Hugo Aragon Franchi

Clover

Drone Diagnostics

Filming at a concert
Credit: D. Vision Aerials

Problem Context

- Sometimes operated in densely populated areas
- Minor motor faults can cause crashes and damage
- Detection relies on subjective operator listening

Opportunity

A portable system to verify the health and flight readiness of drone propulsion systems using objective pre-flight diagnostics.

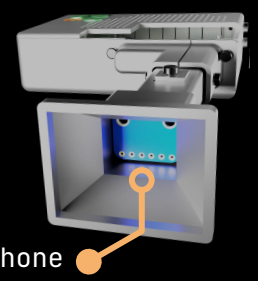
Objectives

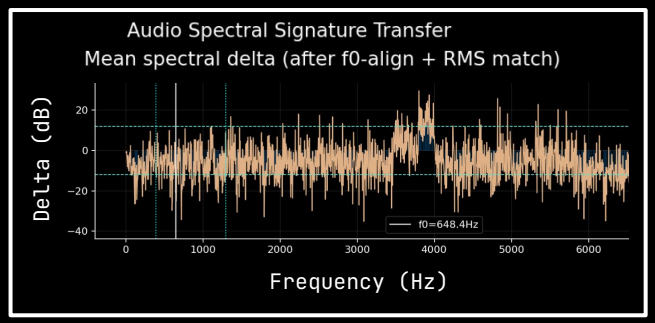
1. Fault Diagnosis
2. Diagnostic Reliability
3. Operational Integration
4. Field Portability
5. Diagnostic Traceability

Clover is a portable drone diagnostic device that analyzes motor acoustics and flight data to assess propulsion health and flight readiness before flight.

[1] Audio Analysis

Start Session → Record Audio → Analyze Audio

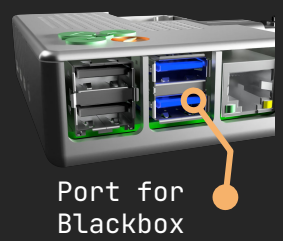


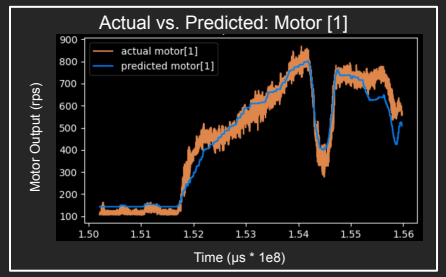


Deep neural network trained on existing dataset. Drone audio is transformed to input audio.

[2] Blackbox

Record Blackbox → Analyze Blackbox





Deep neural network trained on existing dataset, transformed to fit the tested drone.

Tests: Motor output / Setpoint matching

[3] Results

Display Results → Erase Flash → End Session

Result(A): go
Audio: normal
Blackbox: all tests passed

Result(A): no-go
Audio: Anomaly (Motor Fault)
Blackbox: Anomaly (Test 1 failed)

Result(A): uncertain
Neither tests were performed.

Deep neural network trained on existing dataset. Drone audio is transformed to input audio.

[4] Logging

Sync Logs → View Logs